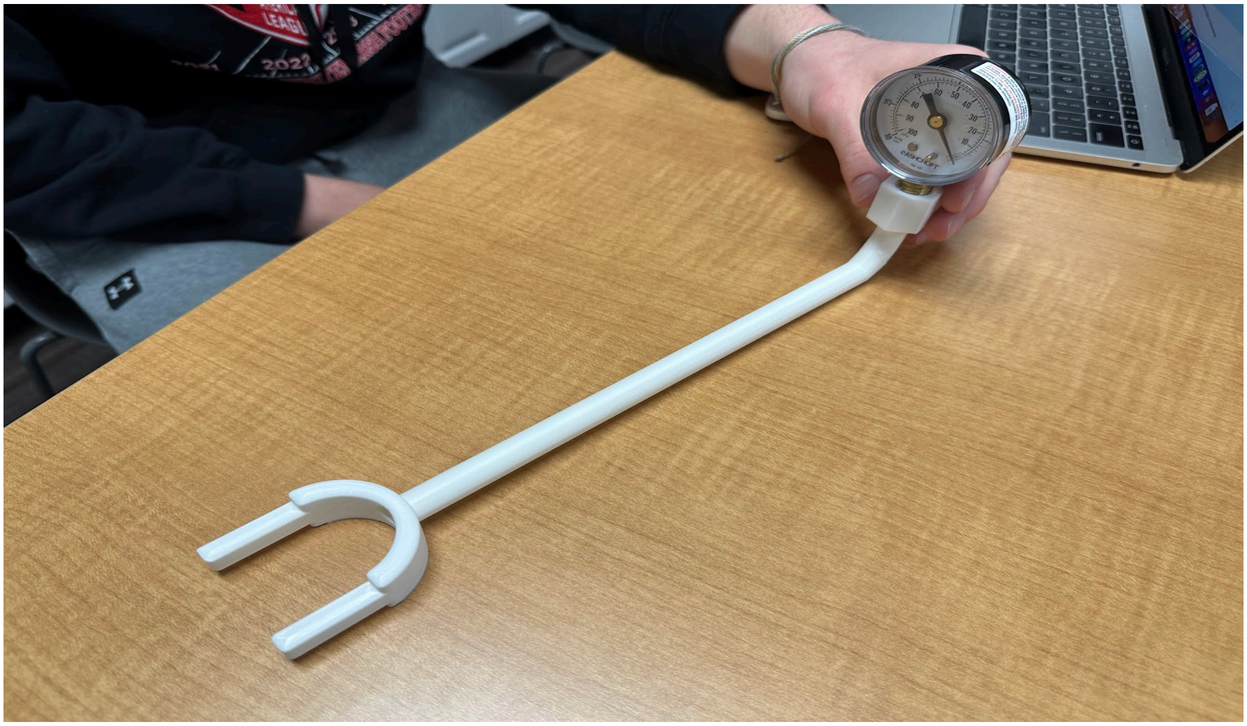
G1. Final prototype iteration explanation:

The final prototype iteration of the Maximal Inspiratory Pressure (MIP) device is compact, durable, and functional. We plan to buy a negative inspiratory pressure measurement tool (a dial) to meet our key requirements. Our design integrates a mechanical pressure gauge with a 3D-printed body, keeping the device lightweight and easy to transport, while also being cost-effective and simple to manufacture.

Key Design Details:

- **Materials:** 3D printer plastic makes up the mouthpiece and connector tube. The pressure gauges are bought from McMasterCarr.
- **Pressure gauge:** We selected a negative mechanical pressure gauge due to its reliability and accuracy. We made sure it works for the average human air pressure, since most on the market were for industrial use. This gauge is compact, durable, and affordable.
- The final prototype iteration includes a better pressure gauge to fit our needs to facilitate easier use by students. These changes reflect feedback from initial testing and expert consultation.
- **3D-printed body:** The mouthpiece and the attachment for the pressure gauge is 3D printed using PLA filament, a material used for its availability and decent durability. The body was designed to be lightweight but strong enough to withstand repeated use without significant degradation.

Photos & Sketches:



This picture demonstrates our prototype. It features a mouthpiece attached to a tube, in which the air travel through. At the end of the tube, the dial is attached to a screw, making a firm connection between the two.

IMG_0861.MOV

This video is the result as the device is being used. As you can see, the dial moves counterclockwise as the inhalation is occurring.

This picture depicts all of the individual parts, emphasizing the simplistic, yet reliable/accurate design.

G2. Construction of the final prototype:

The final prototype has been constructed with enough detail to ensure that objective data on all of the design requirements can be collected effectively. The MIP device has been built to meet the primary design goals of durability, accuracy, simplicity, and portability, and it has undergone modifications to enhance its functionality.

Objective Data Collection:

- **Durability and Longevity:** The prototype's 3D-printed PLA body is designed to endure continuous use in a classroom setting. The material is well-suited to withstand at least five years of repeated testing, with minimal degradation.
- **Accuracy:** The pressure gauge is calibrated to provide measurements with an accuracy of ±1 mmHg, which is within the required tolerance for the MIP test. This accuracy will be verified through a series of controlled tests using known pressure values, ensuring that the device provides consistent and reliable results.

- **Portability:** The compact design of the device allows for easy transportation and storage, which is essential for use in a classroom setting with a large group of students. The lightweight construction ensures that the device can be carried from station to station without difficulty. Further testing will involve transporting the prototype through various setups to confirm its portability.
- **Simplicity and Replicability:** The MIP device is easy to assemble, use, and clean, making it a viable option for multiple devices to be used by a large cohort of students. The simplicity of the design ensures that it can be easily replicated for classroom-wide use.

G3. Testing or modeling of attributes:

All key attributes of the MIP device are addressed and can be tested or modeled mathematically.

Attributes to be tested:

- **Accuracy:** The pressure gauge will be tested against a series of known reference pressures to verify its ±1 mgHg accuracy. We will test this by comparing it to a known air pressure gauge (with the one provided at school) and confirm that our device works.
- **Durability:** To test the prototype's longevity, it will undergo repeated cycles of pressure testing, simulating real-world classroom use. We want to confirm that the device can withstand at least five years of daily use without significant degradation in functionality. So, we will put it in more extreme conditions than it would actually experience, like conducting a couple of drop tests, to find out if it works.
- **Portability:** The prototype will be subjected to multiple transport tests, where we will carry between our different work stations and stored in various environments (e.g., desks, locker) to confirm that it is easy to transport and store. This obviously can't be tested really well unless it is taken in a car, so we will ensure there is proper packing.
- **Ease of Use:** We will ask a couple of classmates to test our device for us to ensure that it is simple to set up, use, and clean. Feedback from the students will help refine any aspects of the device that may require improvement for better usability.

G4. Addressing non-testable attributes:

While the prototype addresses nearly all the key attributes directly, some aspects, such as the **ergonomic design** for extended use, cannot be fully tested through physical or mathematical modeling. Although user testing will provide some feedback, the ideal ergonomic design varies from student to student, and there is no clear mathematical model for optimizing comfort.

Expert Review:

- **Dr. Joseph Sarver** and **Jaimie Dougherty**, both experts in biomedical engineering and functional design, will be further consulted to provide feedback on the ergonomic aspects of the device. Their expertise in human biomechanics and functional design will help assess whether adjustments are needed to improve user comfort or usability.
- Feedback from these experts will be used to make any necessary refinements to the design, ensuring that the device is both comfortable and functional for a diverse group of students.

Instruction Manual

MIP Instruction Manual.pdf